

Supplemental Material

Supporting “Supervised Tumor Attention Transfers Across CT Cohorts, but EGFR Prediction Does Not Exceed Clinical Variables.”

S1 Training configuration

Reported so the experiment can be understood without reading the repository. All values are the locked configuration used for every run in the manuscript.

Parameter	Value
Encoder	BiomedCLIP ViT-B/16, last two transformer blocks unfrozen
Region head	8 learned queries, cross-attention to patch tokens
Optimizer	AdamW
Learning rate	2×10^{-4} , backbone multiplier 1.0
Schedule	OneCycle, warmup ratio 0.05
Weight decay	0.01
Batch size	8, gradient accumulation 2 (effective 16)
Epochs	40 maximum
Early stopping	patience 8, minimum 10 epochs
Gradient clipping	1.0, with non-finite steps skipped
Loss weights	classification 1.0, attention 0.5, generation 0.0
Model selection	composite inner-validation monitor: $1.0 \times \text{EGFR AUC} + 0.5 \times \text{attention-mass lift}$
Sections per volume	16
Crop	$(\text{bbox}_{\text{mm}} + 24) \times 2.5$, 30% positional jitter
Window	center -600 HU, width 1500 HU
Input size	224×224
Hardware	Apple silicon, Metal Performance Shaders backend

The generation loss weight is zero in every reported run. With that term active, folds hit a

marginal overflow in the backward pass on this backend, skip every optimizer step and report an untrained model; the language decoder is therefore present in the code but contributes nothing, and the model is image-only. `gen_loss` is correspondingly omitted from the selection monitor, because monitoring an untrained term selects on noise.

No hyperparameter search was performed. These values were fixed before the runs reported here and were not tuned against any reported endpoint.

S2 Participant flow and selection

S2.1 Exclusions, with reasons

Stage	<i>n</i>	Reason for loss
Clinical sheet	211	—
With an imaging series	188	23 had no CT series in the collection
Successfully preprocessed	158	30 had no usable tumor localization: no segmentation and no annotation point, a point unresolvable to a section, or a point landing in air (< -700 HU)
Known EGFR call	153	5 recorded as Unknown or Not collected
Adenocarcinoma	133	20 squamous or not otherwise specified
With a segmentation	97	36 had an annotated point but no pixel mask

S2.2 Analyzed versus excluded patients

The 58 patients not analyzed are similar to the 153 analyzed on age, sex and smoking, and differ materially on histology.

Characteristic	Analyzed ($n = 153$)	Excluded ($n = 58$)
Age (y), mean $\pm$ SD	68.8 $\pm$ 8.8	65.9 $\pm$ 12.5
Pack-years, mean $\pm$ SD	40.5 $\pm$ 26.4	37.0 $\pm$ 25.9
Male	64.1%	63.8%
Never-smoker	21.6%	25.9%
Adenocarcinoma	86.9%	67.2%
Squamous cell carcinoma	11.1%	31.0%

This is a real selection effect and we do not minimize it. Exclusion removed squamous patients at nearly three times the rate of the analyzed cohort. Because every squamous patient in this collection is EGFR wild-type, the exclusions are enriched for patients whose genotype is inferable from histology alone. The direction of that bias is to make the analyzed cohort harder than the full collection, so it does not inflate the reported discrimination — but a study reporting higher performance on a cohort selected this way would be partly reporting its selection.

S3 The six analytic choices

The manuscript states that the radiomics comparator moves 0.136 in AUC across six defensible analytic choices. Those choices are two feature definitions crossed with three regularization penalties, on identical patients and folds. The clinical model is shown under the same three penalties for comparison: it moves 0.010.

Features	Penalty	AUC	95% CI
Legacy, 18 features	L2	0.662	0.568, 0.754
Legacy, 18 features	L1	0.605	0.501, 0.706
Legacy, 18 features	L1 + L2	0.604	0.498, 0.707
Rich, 61 features	L2	0.604	0.497, 0.707
Rich, 61 features	L1	0.526	0.416, 0.631
Rich, 61 features	L1 + L2	0.568	0.457, 0.671
Radiomics span		0.136	
Clinical, 5 variables	L2	0.774	0.677, 0.859
Clinical, 5 variables	L1	0.773	0.675, 0.861
Clinical, 5 variables	L1 + L2	0.764	0.665, 0.853
Clinical span		0.010	

A separate comparison holds features and estimator fixed and varies only the validation protocol: flat versus nested cross-validation moves the same arm by 0.066. Neither of these is detectable from an internal point estimate, which is the point.

S4 Reference standard

EGFR status is distributed with the collection and was determined by sequencing of surgical or biopsy tissue by the source investigators. The collection does not publish the sequencing platform, the assay panel, the per-patient tissue source, or the variant-level calls, and we did not have access to them. We therefore report status as the binary label the collection provides and state here that the assay, panel and specimen type could not be harmonized or verified per patient. Any study using this collection is subject to the same limit, and one consequence is that a variant-level analysis — separating canonical activating variants from exon-20 insertions or passenger mutations — is not possible on these data.

The interval from index test to reference standard has a median of 38 days (IQR, 19.5–65.5). No accrual window is recoverable: the collection’s CT dates span 1989–1996 and are de-identification

offsets, so a range derived from them would misdescribe the cohort. The interval survives the offset because it is a difference of two offset dates.

The reference standard for attention is the expert tumor segmentation distributed with each collection, one contour per lesion. Interreader agreement therefore cannot be computed. A mask perturbation sweep (dilation, erosion, translation) is reported as robustness and is not a substitute for a second reader.

S5 Multiplicity and endpoint thresholds

S5.1 The two co-primary endpoints

H_1 (external attention alignment) and H_2 (incremental AUC over the clinical model) test different constructs on different populations — 420 external patients and 153 internal patients respectively — and were both declared before external evaluation. Each was evaluated at $\alpha = .05$ without a gatekeeping rule between them, which is the design actually run and is stated here rather than reconstructed as a hierarchy after the fact. Because H_2 was pre-declared as an expected null and the observed difference is negative, no multiplicity adjustment between the two endpoints would change any conclusion in the manuscript.

Holm correction is applied within the ten-arm pooling sweep, where it does change a conclusion: the best arm moves from $P_{\text{raw}} = .03$ to $P_{\text{Holm}} = .28$, and the peritumoral result is reported as suggestive rather than established because of it.

S5.2 Origin of the 71.4% fold-rate threshold

H_1 was pre-specified as “beats the per-fold spatial shuffle in at least 10 of 14 evaluations,” a count taken from the fold total of the internal run set at the time the plan was fixed. The number of external evaluations is however many run $\times$ fold checkpoints exist, which for the canonical three runs is 15. The count was therefore expressed as the same proportion, $10/14 = 71.4\%$, to preserve the intent rather than silently gaining or losing a required fold. This is recorded as a deviation in the analysis plan’s deviation log, and the observed 11 of 15 clears it by one evaluation.

S5.3 What “pre-specified” does and does not mean here

The plan, including both endpoints and all controls, was fixed and version-controlled before external evaluation. It was not fixed before the external data existed: the cohort was on disk, and the metric’s qualitative behavior on it partly known, when the threshold was chosen. H_1 is therefore a locked evaluation following preliminary development, not a blind confirmatory external validation, and we describe it that way. The distinction matters and is easy to blur, so it is stated in the manuscript’s limitations as well as here.